6

Principles of management of the badly broken down tooth

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6.1 Causes of broken down teeth

This textbook has covered the common causes of broken down teeth: dental caries, tooth wear, and trauma. In addition, long-term failure of parts, or all, of the existing tooth–restoration complex can be significant and may require further operative intervention for its successful management (see Chapter 9). Many intra-coronal defects can be repaired with direct adhesive restorations, as discussed in Chapters 5 and 9. However, the situation can be complicated by the loss of significant portions of existing restoration or tooth structure (e.g. cusps, buccal/lingual walls), which influence the restorative procedures used in an attempt to maintain the tooth longevity, as well as pulp viability, for as long as possible. For direct restorations to succeed clinically, they require healthy dental tissues to aid support, retention, and ideally provide an element of protection from excessive occlusal loads. With diminishing amounts of tooth structure to work with, greater thought and care are required to manage and prepare the remaining viable hard tissues to support and retain the larger restoration.

The *core restoration* describes the often large direct plastic restoration used to build up the clinically broken down crown. It is retained and supported by remaining tooth structure wherever possible (sometimes including the pulp chamber and posts in root canals of endodontically treated teeth). These large restorations often benefit from further overlying protection to secure their clinical longevity, by means of *indirect onlays,* and *partial or full coverage crowns.*

6.2 Clinical assessment of broken down teeth

Before carrying out a detailed clinical examination of the individual tooth and the related oral cavity, it is always important to *justify* your clinical decisions, for both operative and non-operative preventive interventions.

6.2.1 Why restore the broken down (or any) tooth?

The five key reasons for minimally invasive (MI) operative intervention are:

- to repair hard tissue damage/cavitation caused by the active, progressing caries/tooth-wear process (where non-operative prevention has failed repeatedly)
- to remove plaque stagnation areas within cavities/defects which will increase the risk of caries activity due to the lack of effective plaque removal by the patient
- to help to manage acute pulpitic pain caused by active caries by removing the bacterial biomass and sealing the defect, thereby protecting the pulp
- to restore the tooth to maintain structure and function in the dental arch
- aesthetics.

6.2.2 Is the broken down tooth restorable?

This is a very important question to answer pragmatically for the patient sitting in the dental chair at the time of consultation. It is important to manage patient expectations in this regard, as often the operative treatment required is complex technically, takes time over multiple appointments, and may be expensive. Critical issues will include the longevity and functionality of the final restoration; the patient has to appreciate the cost weighed against the clinical benefit of the treatment proposed, along with any significant risks of short-, medium-, or long-term failure. The factors that must be considered when deciding whether to restore a broken down tooth are outlined in Table 6.1. These principles must, of course, be applied to any teeth that are being restored, but are highlighted again in this chapter.

The clinical oral and dental assessment follows the guidance outlined and discussed in Chapters 2 and 3. After obtaining a comprehensive patient history and performing a clinical examination of the whole mouth, the following tooth-related factors should be determined:

- periapical radiographs to help to ascertain the periapical status, presence/absence/quality of root canal treatment (RCT), extent of coronal tissue damage relative to the pulp chamber, root/canal morphology, and alveolar bone levels
- pulp sensibility/vitality
- an occlusal analysis to ascertain the loading, guidance, or interferences that may be related to the tooth in question, in all axes of movement
- occlusal vertical dimension (OVD) assessment to ensure that there is enough inter-occlusal space to place a restoration successfully. Tilting/over-eruption of the tooth or opposing teeth may reduce the space available in which to place a secure restoration with adequate strength, so forcing treatment down the more complex reorganized approach to dental rehabilitation (see Chapter 5, Section 5.14).

When considering the tooth to be restored itself, further factors to consider would include the following:

- *Amount/distribution of the remaining viable tooth structure.* In posterior teeth, the presence of solid cusps directly or diagonally opposite one another can help to support and even retain the core restoration with relative undercuts (see later). On the other hand, thin, weaker retained buccal or lingual/palatal walls would be less amenable to preparation for the retention of amalgam, and more prone to fracture. In these situations, the retained, weaker cusps/walls may be reduced in height vertically and the core material overlaid to offer some occlusal loading pressure relief during function (see Figure 6.1). When placing adhesive resin composite cores, care must be taken to limit shrinkage stresses at the tooth–restoration interface using incremental techniques and when considering the axial tooth preparation that might be required for the indirect crown to be fitted.
- *The sub-gingival extent of the coronal damage.* When fragments of tooth–restoration complex have been lost in function, the fractures

Table 6.1 General clinical factors to be considered when deciding whether a broken down tooth is restorable or not in a particular patient

Factor	Favourable	Less favourable
Pulp	Vital pulp with adequate RDT to incorporate tooth preparation	Symptoms present, PA pathology (RCT will be required; isolation/access cavity compromised?)
RCT status	Good-quality sealed RCT, assessed radiographically with PA resolution evident. Straight canals and broad roots facilitate post placement	Symptomatic, recurrence of PA pathology, re-RCT required? Thin roots and curved canals prevent post placement
Tooth structure	Adequate bulk remaining (e.g. cusps) Adequate distribution of coronal tissues. No cracks/defects	Inadequate quantities available once tooth preparation has occurred leads to weakness and fracture
Existing restoration status	Only requires refurbishment or repair (see Chapter 9)	Active caries present, leaking margins, subgingival extension
Periodontal status	Pocket depths 3–4 mm, minimal recession, > 50% bone level from PA radiographs	Active periodontal disease, LOA, < 50% bone levels, grade II–III mobility
Occlusion Dental arch tooth position	Protected canine guided lateral excursions. No interferences Fully/partially dentate with even tooth distribution, minimal tilting/rotations	Reduced OVD minimizes space for restoration Tooth bears full occlusal load in a quadrant, drifted/tilted/rotated
Aesthetics	Posterior dentition, previously restored dentition	High smile line in anterior aesthetic zone, recession risk due to thin gingival tissues, subgingival restoration margins, and poor contour; complex shade matching
Patient factors: preventive status	Well-motivated responsible patient, efficacious oral hygiene control (or can be educated to be such)	Refuses to take responsibility for/unable to maintain oral health; poor diet, oral hygiene, no fluoride use

RDT, remaining dentine thickness; RCT, root canal treatment; PA, periapical; LOA, loss of attachment; OVD, occlusal vertical dimension.

that occur often follow natural internal histological planes, leading to shear fractures that can terminate on the root surface several millimetres subgingivally (see Figure 6.2). This makes their restoration difficult in terms of preparing the subsequent base cavity form to support the core restoration as well as gaining direct access and moisture control, thus adversely affecting the successful medium- to long-term restorability of the tooth.

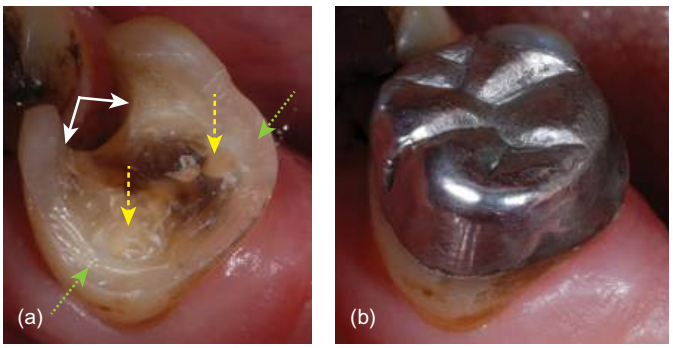


Figure 6.1 **(a)** A heavily broken down mandibular molar. The original mesial-occlusal-distal (MOD) amalgam cavity has been modified macromechanically to aid retention and support of the larger core restoration to be placed. The white arrows show the relative proximal undercut in the distal box. The green dotted arrows indicate how the thinner, weaker mesiobuccal cusp and lingual walls have been reduced in occlusal height and prepared with a shoulder finish to support the new overlaid amalgam restoration margin (and be protected by it). The yellow dashed arrows show retentive grooves which have been cut into the healthy dentine (avoiding the pulp chamber), into which amalgam will be packed. **(b)** The large amalgam core restoration after several years of function. (Courtesy of L. Mackenzie.)



Figure 6.2 A mirror image of the UR5, 6, 7, 8, showing a traumatic distal shear fracture of the still vital UR7 terminating subgingivally on the root surface.

Q6.1: What practical difficulties might be encountered when trying to restore the tooth shown in Figure 6.2?

- *Using the pulp and/or root canal spaces.* If the tooth has undergone successful root canal treatment (RCT), there is the potential to prepare the straight portion of a root canal for a post, which can then help to retain the core restoration (see Figure 6.4). Restorative core material can fill the pulp chamber and even the first 2–3 mm of root canal space, close to the orifice, so gaining mechanical retention. When this type of restoration, originally using amalgam, is placed, it is known as a 'Nayyar core' (see Chapter 7, Section 7.5.3 and Chapter 8, Section 8.10). With the development of modern, high-quality adhesive dental restorative materials to aid core retention, the need for elective RCT of a vital tooth for core retention is not normally indicated.
- *The direct core restorative material.* The traditional restorative material used for core restorations, especially on posterior teeth, is dental amalgam, due to its high compressive strength and wear resistance under occlusal loading. Contemporary resin composites have an improved strength and wear resistance compared with their predecessors, and as they will often be covered with an indirect crown, these are now becoming more popular as core materials (see Figure 6.4e). Their adhesive retention properties reduce the amount of tooth preparation required, making them an attractive proposition where remaining tooth structure is at a premium (see Chapter 5, Section 5.10). However, moisture control and the relative complexity of their placement technique in multiple increments in cavities that are not always easy to access operatively or keep dry can limit their use successfully in this regard.

6.3 Intra-coronal core restoration

The key functions of the direct plastic core restoration used to build up the broken down tooth include:

- reconstituting the structure, function, and possibly the aesthetics of the crown of the tooth
- maintaining pulp vitality, or where a RCT is present, ensuring that the coronal endodontic seal is maintained
- providing indirect support, stability, and retention for the overlying indirect laboratory-made restoration, if required
- de-bulking/reducing the volume of the more expensive overlying indirect restoration—this will be more relevant when precious metals are used in its construction.

The direct core restoration is in the form of a plastic restorative material (e.g. amalgam, resin composite). However, in some cases, depending on the distribution and quantity of remaining tooth structure and the size and shape of the existing cavity, an indirect/cast inlay/onlay or partial crown restoration may be placed instead of the plastic core build-up. Materials used for this type of restoration would include indirect resin composite, precious and non-precious metal alloys, or ceramics.

6.3.1 Direct core retention

The retention mechanisms used for different plastic restorative materials have been highlighted in Table 5.8 in Chapter 5. *Macro-mechanical* cavity modifications to aid restoration retention will include the provision of diametrically opposing hard tissue undercuts where possible, without over-preparing and weakening the remaining tooth structure unnecessarily. Opposing surfaces can be made more plane parallel with marginal shoulder preparations to stabilize the final restoration, and minimally invasive stabilizing slots and grooves can be placed within healthy tissues where possible (see Figure 6.3).

C factor

Adhesive materials will be placed with suitable bonding systems (see Table 8.12 in Chapter 8). Incremental layering techniques must be used for conventional resin composite materials. This minimizes the polymerization shrinkage stresses concentrating at the tooth–restoration interface which can lead to potential tooth flexure and even cracks, with concomitant symptoms of sensitivity. However, in large open cavities with a reduced ratio of bonded surface area to free surface area (the C factor), the shrinkage stresses will be less problematic.

- An occlusal cavity with a ratio of five bonded surfaces to one unbonded surface has the highest C factor, of 5 (i.e. 5:1), thus maximizing the interfacial stresses.
- A labial resin composite veneer has the lowest C factor, of 1 (a ratio of one bonded to one free surface, both of similar surface area). In this case, the shrinkage (on average in the range 1.5–2.5% by volume for many modern resin composites) will be dissipated within the restoration and not concentrated at the tooth–restoration interface (see Figures 6.4d and 6.4e).

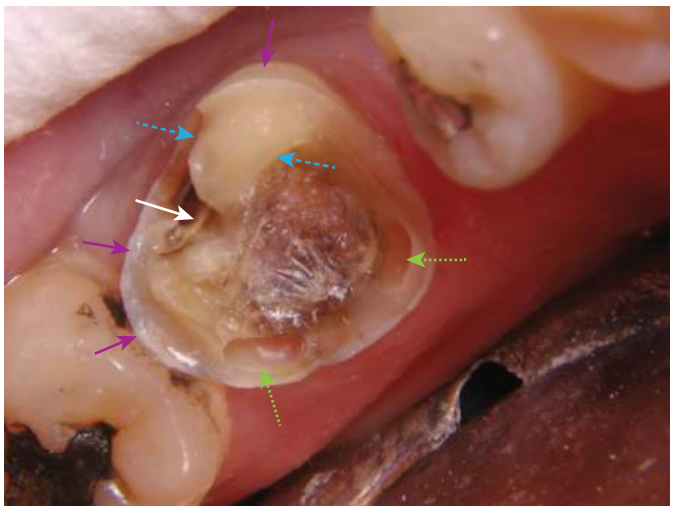


Figure 6.3 A heavily broken down LL6 which has been prepared macro-mechanically to receive an amalgam core restoration. The blue dashed arrows indicate the relative undercut prepared around the vertically reduced mesiobuccal cusp, which will be overlaid with amalgam. The green dotted arrows point to two slots that have been prepared in healthy dentine to aid retention of the amalgam. The small purple arrows highlight the prepared distal-buccal shoulder to support the amalgam restoration margin. The amalgam placement procedure is described in detail in Chapter 8, Section 8.9. (Courtesy of G. Palmer.)



Figure 6.4 **(a)** Fractured UR1. The crown fractured through the pulp chamber during mastication. The remaining root was endodontically treated and a temporary restoration was placed in the access cavity. The fracture margins are just at the gingival margin or supragingival. The remaining enamel and dentine are sound. **(b)** and **(c)** show the root canal post space having been created using increasing sizes of Gate Glidden burs. A fibre-reinforced direct post has been placed into the post space. Its length has been adjusted to ensure that it is short of the incisal edge position of the final restoration. **(d)** The fibre-reinforced post has been cemented into the canal space using a resin cement, and the excess has been removed before final photo-curing. **(e)** Direct provisional resin composite restoration built up incrementally around the fibre post. **(f)** Definitive indirect ceramic cast restoration UR1 in occlusion. **(g)** Final periapical radiograph showing the fibre-reinforced post in place with 4 mm of root canal filling material (gutta-percha) still present at the apical end of the root canal. Note the similar radiodensity of the fibre post compared with the surrounding dentine.

Q6.2:

- **i** Look at Figure 6.4 (a). What other restorative materials could be used to fill the endodontic access cavity?
- **ii** Look at Figure 6.4 (b). Why should the post diameter not exceed one-third of the root diameter?
- **iii** Look at Figure 6.4 (c). Why should the post extend to 1–2 mm below the incisal edge/occlusal surface of the definitive restoration?
- **iv** Look at Figure 6.4 (d). Can you suggest any alternative materials used to cement a root canal post?
- **v** Look at Figure 6.4 (e). What is the C factor of this provisional restoration?
- **vi** Look at Figure 6.4 (f). Why should the post length within the root canal exceed the length of the final restoration it supports?

To date, there is limited independent published evidence to validate the use of bulk-fill resin composites, which only need one large increment of material to fill a larger cavity, as a core restorative material (see Chapter 7, Section 7.2.2). However, this class of resin composite is popular among dentists due to its simplicity of use and—anecdotally at present—these materials seem to offer at least medium-term success. High-viscosity glass ionomer cements and their derivatives have been used as core materials, but as a general rule these materials are not recommended routinely for this purpose unless there are at least two opposing walls of tooth tissue present. Their only moderate compressive strength and wear resistance, when compared with amalgam and resin composites, also makes them less appropriate for use as long-lasting restorations exposed to extensive occlusal functional loading.

In the restoration of endodontically treated teeth, the cementation of a post into the root canal space is indicated if the amount of residual tooth structure is insufficient to support a core made of a plastic material (amalgam or resin composite). The roots must also have sufficient bulk (diameter), and must have a length of straight root canal longer than the height of the final restoration it will support. Root canal posts require removal of the root canal filling material (often gutta-percha) within the coronal two-thirds of a straight canal. The procedure usually commences with a small rose head bur in a slow-speed handpiece, then continuing with a specially designed steel Gates Glidden bur. This consists of a non-cutting point on the end, which melts the gutta-percha, and then a bulbous cutting tip with a narrow shank behind that allows the root filling material to be spun out of the canal space—all in one action. The coronal root canal space then needs to be increased in size to accommodate a post. This is achieved with a special twist drill (often supplied with the post kit) mounted in the slow-speed handpiece, shaped to match the size of post selected. The maximum width of the post is normally equal to one-third of the diameter of the remaining root, assessed from the periapical radiograph. There should be at least 4 mm of residual root canal filling material remaining at the root apex, to ensure that the existing apical endodontic seal is not disturbed. All of these dimensions can be assessed from the preoperative periapical radiograph (see Figure 6.4).

6.4 Clinical operative tips

- The ability to achieve successful moisture control is critical to the ultimate long-term restorability of the broken down tooth. Clamping the individual tooth for rubber dam placement may not be possible due to the scarcity of supra-gingival tooth structure. A split-dam technique may be required, with the dam anchored to adjacent teeth and the use of circumferential floss ligatures around the broken down tooth, to adapt the dam to the tooth to be restored (see Section 5.6.3).
- In some cases, the use of cotton wool rolls/judicious suction may be all that is possible in this regard (see Section 5.6.2).
- Historically, a suitably adapted copper ring or, if available, an orthodontic band can be used to support the core restoration build-up circumferentially, and retained while the restorative material sets in function, before its removal at a subsequent visit.
- Attempt to make internal cavity walls as plane parallel or slightly undercut as required without the concomitant loss of unnecessary sound tooth structure. Opposing walls help the stability of the core restoration within the cavity (see Figures 6.1a and 6.4).
- Careful use of slots, grooves, and pits can help the retention of large core restorations (see Figure 6.1a and 6.4).
- *Matrixing:* Various designs of circumferential matrix exist, ranging from copper/orthodontic bands to self-tightening and supporting auto-matrix systems (see Chapter 5, Section 5.12). These are most useful for aiding the build-up of multiple missing walls of large cavities. Problems commonly arise when there is little peripheral tooth structure to support the outline shape of the band and, on tightening, the band collapses over, especially where extensive cusps and walls are missing, resulting in open proximal contacts and poor final restoration morphology.

  Clinical techniques to overcome this loss of restoration morphology and contact points include the following:



Figure 6.5 A mirror view of a stainless steel orthodontic band placed around the LL6, supporting a large coronal restoration following endodontic treatment and prior to provision of an indirect full-coverage restoration. Note how the band has been adapted well to the contour of the molar and burnished into the furcation defect buccally. Regarding the clinical photography, note the use of cheek retractors and an occlusal mirror which is kept demisted by first heating it up in warm water prior to drying it and using it intra-orally. It is then subjected to a gentle air stream from the 3-1 syringe by the nurse who holds the mirror steady. The patient can hold the retractors once positioned by the operator.

Q6.3: What clinical clues in the patient shown in Figure 6.5 might help you gain an insight into the patient's high caries risk, prior to when the image was taken?

- Trimming/shaping the outline of the matrix band to help proximal placement through tight contact points, adaptation, and contour of the restoration.
- Using a ball burnisher hand instrument (see Figure 5.26 in Chapter 5), burnish the matrix band once in place, tightened and wedged proximally to adapt the band to the cervical margin of the tooth.
- Tightening and wedging the band proximally, filling the depth of the proximal boxes with initial small increments of restorative material (e.g. flowable resin composite). Then loosening the matrix a little to adjust its position and allow burnishing adaptation at the contact point, followed by further increments of material. This 'adaptive matrixing' permits close adaptation at the cavity floor and adequate contouring of the proximal restoration anatomy.
- Initially restore the accessible cusps, lingual/palatal, and buccal/labial walls without the aid of a matrix band. Once restored, place the matrix band conventionally (circumferential or sectional), wedge and restore the proximal areas to the correct contact point and proximal morphology.
- The clinical procedure for placing a fibre-reinforced post and building up a resin composite core is outlined in Table 8.11 in Chapter 8. Endodontically restored teeth may need endodontic re-treatments, and in this respect well-bonded fibre posts can be removed, but considerable care is needed. Their removal is achieved by drilling down through the middle of the post, avoiding any additional removal of peripheral root or coronal dentine. In contrast, removal of a non-adhesively cemented metal-based post may be relatively easy, using an ultrasonic scaler to break down the cement at the post–dentine interface. Unfortunately, an adhesively cemented metal post will be nearly impossible to remove without damaging the remaining root dentine.

6.5 Design principles for indirect restorations

This section aims to *outline* the common design features required for the successful placement of indirect laboratory or chair-side made restorations used to reconstruct the crowns of broken down teeth. It does not cover the details of specific individual step-by-step restoration preparations, materials, or operative techniques involved in the clinical and laboratory manufacturing process. This information is available in alternative prosthodontic texts.

Indirect restorations are classified as those that are fabricated outside the mouth—either by dentists making temporary crowns from putty indices, or technicians making restorations in a dental laboratory or from a clinical CAD-CAM processing unit from intra-oral records—and then cemented into place clinically by the operator (see Table 6.2). They may be appropriate in the following clinical situations:

- to protect a broken down, weakened tooth from further fracture or to hold together parts of a cracked tooth (cusps, walls)
- to restore an already broken tooth or a tooth that has been severely worn down
- to cover, support, and protect a tooth with an extensive core restoration from large occlusal loads when there is little tooth structure remaining
- to retain a bridge, whose *pontic(s)* replace the missing tooth/teeth and *abutments* anchor the prosthesis to adjacent tooth/teeth
- to cover severely misshapen or discoloured teeth where direct restorative methods would not be clinically or practically viable

Table 6.2 The different designs of indirect restorations and the range of materials used to manufacture them. The intrinsic properties of the materials, often needing to be optimal in thin section, will include a combination of suitable compressive, shear, and tensile strengths (to combat occlusal loading), wear resistance (ideally comparable to enamel), and aesthetics

Indirect restoration type	Materials
Inlay	Gold type I alloy, ceramic, laboratory composite
Onlay	Gold type II and III alloy, composite, ceramic
Partial coverage crown (e.g. three-quarter crown)	Gold type III alloy
Full coverage crown (full gold crown (FGC), metal ceramic crown (MCC)/porcelain-fused-to-metal crown (PFM), all-ceramic crowns)	Gold type III alloy, palladium alloy, base metal alloys (nickel, chromium), ceramic, stainless steel (paediatric crowns)
Precision attachments	Gold type III and IV alloy, palladium or base metal alloy (nickel, chromium)
Temporary crowns	Acrylate resin/acrylic based

- to restore dental implant abutments and to act as precision attachments for removable prostheses.

For children, an indirect crown may be used on the primary dentition in order to:

- save a tooth that has been so damaged by caries that it can longer support a filling, but requires maintenance in the dental arch for functional and developmental reasons (the Hall technique)
- protect the teeth of a child at high risk for caries, especially if he or she has difficulty maintaining daily oral hygiene
- decrease the frequency of sedation and general anaesthesia for children who are unable because of age, behaviour, or medical history to fully cooperate with the requirements of proper dental care/maintenance.

In such cases, a paediatric dentist is likely to recommend a stainless steel crown.

6.5.1 Design features

Retention is the property of a restoration/cavity that prevents its displacement from the cavity in its direction of placement/insertion. Prior to the use of chemical adhesives, this property was reliant on macromechanical tooth preparation features, including undercuts (only for direct plastic restorations), slots, and grooves, which potentially weakened and sacrificed quantities of sound tooth structure (see Chapter 5, Section 5.10).

Resistance form is the property of a restoration/cavity that prevents its displacement in any other direction.

These traditional terms have gone out of fashion somewhat, and indeed their use was always rather more theoretical than practical. Their relevance is much reduced in contemporary minimally invasive operative dentistry since the advent and development of dental adhesives. However, basic mechanical principles are still relevant and important when designing preparations suitable for indirect restorations of all types of material:

- *Occlusal/axial reduction.* This should be sufficient to accommodate the appropriate thickness of restorative material used for the indirect restoration. Gold alloy crowns require the least tooth preparation, as they have adequate strength/ rigidity in thin section (0.7 mm). Indirect resin composites and all-ceramic restorations require a greater bulk (up to 2 mm) to achieve this. Newer ceramic formulations allow thinner veneers with adequate strength and aesthetics, so making tooth preparation more minimally invasive.
- *Axial wall taper* (see Figure 6.6*a and b*). For indirect restorations that have been manufactured and cast or cured outside the mouth, perfectly parallel walls would be ideal, but in practice these are impossible to achieve without introducing an element of undercut. Instead, a slightly divergent axial taper of 7–10 degrees is recommended. Even this level of taper is difficult to achieve/perceive with the naked eye in the oral cavity; therefore 15 degrees is often the minimum achieved in practice. Any undercut in the axial surface preparation (as with classic amalgam cavities, see Chapter 5, Section 5.10, Figures 5.48 and 5.49) is contraindicated, as it will be impossible to make or seat the restoration.
- *Surface area and finish.* Maximizing the surface area between the indirect restoration and tooth will aid cementation and therefore



Figure 6.6 **(a)** A maxillary anterior occlusal view of full coverage indirect crown preparations, UR3-UL3. Note the supragingival location of the rounded shoulder finishing margins of the tapered axial walls. **(b)** The labial view of the same tooth preparations where the rounded shoulder finishing margins are more clearly demarcated, as is their circumferential position supragingival to the surrounding gingival papillae. The relative tapering of the axial walls can be clearly seen on the upper central incisor teeth. Sufficient occlusal height reduction of the teeth allows **(c)** the final ceramic restorations to be cemented using a resin-based luting cement. (Courtesy of Dr Ryan Olley.)

Q6.4:

i What are the advantages and disadvantages of placing the axial wall finishing margins supragingivally as shown in Figure 6.6 (a)?

ii Do you know what the black lines are adjacent to the finishing margins of the six prepared teeth as shown in Figure 6.6 (b)?

retention. The microscopic surface produced by fine dental diamond/tungsten carbon finishing burs will increase the surface area for adhesive bonding without introducing major flaws or cracks in the tooth tissue (see Chapter 5, Sections 5.82 and 5.83).

- *Preparation margin (see Figure 6.6a and b): position*. To maximize the axial wall length and surface area, it is desirable to place the finishing margin of the full coverage crown close to but just supragingival. This will aid oral hygiene procedures for the patient and allow simpler impression taking for the operator. Care must be taken during tooth preparation not to traumatize the gingival tissues or to encroach on the biologic width of the gingival attachment complex (the region between the alveolar bone crest and junctional epithelial attachment), for fear of accelerating irreversible periodontal tissue damage and recession. The patient must be advised about optimal oral hygiene procedures to prevent plaque biofilm stagnation at the restoration margins. Indeed, if a patient shows evidence of not being able to maintain suitable oral hygiene around such areas, this should potentially act as a contraindication to restoration placement (see Table 6.1).
- *Preparation margin: shape*. The cross-sectional shape of the margin between indirect restoration and tooth should be a *chamfer* or a *rounded shoulder* configuration. This is dependent on the following:
  - The type of material used to fabricate the restoration. Gold and other metal alloy margins can be manufactured to finer tolerances, so reducing the amount of tooth preparation required at the interface between restoration and tooth. Ceramics require a greater bulk of material for strength and aesthetics, resulting in heavier marginal tooth preparation (see Figures 6.6c and 6.7).
  - The amount of tooth structure/core restoration present. The most tooth-preserving, minimally invasive margin preparation is a minimal chamfered finish, which may be desirable where remaining tooth structure is at a premium and not easily accessible.
  - The position of the margin. When lingual, distal, or palatal, margins can be finished in metal without an aesthetic compromise, so needing only a minimally invasive chamfer finish. Indeed, circumferential indirect restoration margins may have different configurations dependent upon their anatomical position (labial vs. proximal vs. lingual/palatal).

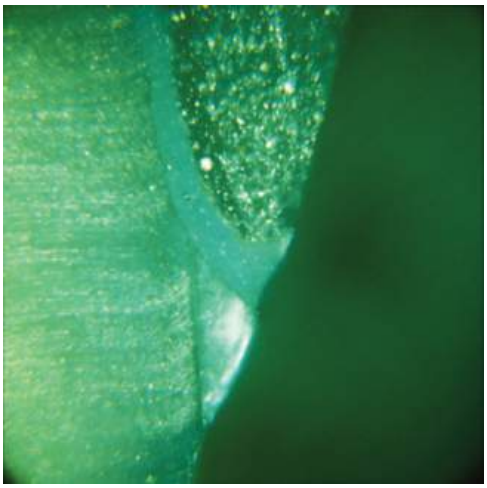


Figure 6.7 A photomicrograph (field width 2 mm) showing a section through the margin interface of a thin ceramic restoration luted to a tooth. The dentine is on the left of the image, and the ceramic restoration is on the upper right. Between it and the dentine and the small triangle of retained enamel is the resin-based luting cement. The profile of the axial finishing margin on the tooth is a chamfer.

The different margin configurations are prepared using end-cutting burs with a rounded tip (half depth) to achieve a chamfer (see Figure 6.7), and a squared off tip profile to achieve a shoulder.

- *Adhesive resin cement*. Modern adhesive cements have chemical moieties that can bond to ions in the tooth as well as oxides and other chemical groups within the restorative material, so providing some chemical adhesion (see Figure 6.7). This permits greater leeway in the mechanical accuracy of the tooth preparation (including the degree of taper of the axial walls, their surface finish, and surface area).

6.6 Answers to self-test questions

Q6.1: What practical difficulties might be encountered when trying to restore the tooth shown in Figure 6.2?

A:

i Difficult physical instrument access to the distal of the UR7.

ii Difficult moisture control (hard to place rubber dam).

iii Difficult to modify cavity form at the base of the shear fracture.

iv Difficult to place restorative material at base of fracture (adhesive or otherwise). A possible solution to improving the restorability of the tooth would be to carry out a crown-lengthening procedure to relocate the gingival margin apically, so improving access to the base of the cavity.

Q6.2:

i Look at Figure 6.4 (a).What other restorative materials could be used to fill the endodontic access cavity?

A: GIC, resin composite.

ii Look at Figure 6.4 (b). Why should the post diameter not exceed one-third of the root diameter?

A: If the post is wider than this ratio, there is an increased risk of root fracture, especially with metal-based posts which do not dissipate stresses evenly through the root length.

iii Look at Figure 6.4 (c). Why should the post extend to 1–2 mm below the incisal edge/occlusal surface of the definitive restoration?

A: So that there is enough space afforded to the core restoration to cover the post. When the core restoration is prepared for a crown, up to 2 mm of clearance may be needed occlusally/incisally. There is a risk that the post may be exposed during crown preparation, and this can potentially weaken the core structure or even the fibre post.

iv Look at Figure 6.4 (d). Can you suggest any alternative materials used to cement a root canal post?

A: GIC-based cements.

v Look at Figure 6.4 (e). What is the C factor of this provisional restoration?

A: The C factor is calculated as the ratio of the bonded surface area to the dentine vs. the non-bonded surface of the resin composite restoration. As you can see from the series of images, the bonded surface is small compared with the rest of the restoration, so this is a favourable configuration. Thus stress concentrations at the interface will be minimal.

vi Look at Figure 6.4 (g). Why should the post length within the root canal exceed the length of the final restoration it supports?

A: Mechanically, this provides the best distribution of occlusal loading forces on the post–restoration–tooth complex. If the post length is shorter than its supported restoration, unfavourable fulcrums of force can be generated, leading to crown or root fracture in clinical use.

Q6.3: What clinical clues in the patient shown in Figure 6.5 might help you gain an insight into the patient's high caries risk, prior to when the image was taken?

A: The presence, site, and number of tooth-coloured adhesive restorations and the apparently untreated buccal carious lesion on the LR5.

Q6.4:

i What are the advantages and disadvantages of placing the axial wall finishing margins supragingivally as shown in Figure 6.6 (a)?

A: Advantages: no trauma to the gingiva during preparation; easier recording of margins in the final impression; easier to construct accurate temporary crowns; easier for the laboratory to fabricate accurately fitting crowns; less likelihood of the periodontium being adversely affected and thereby leading to later recession or periodontal disease; easier for the patient to clean the restoration margins. Disadvantages: anteriorly, aesthetics may be compromised as the margins may be visible in a patient with a high lip line.

ii Do you know what the black lines are adjacent to the finishing margins of the six prepared teeth as shown in Figure 6.6 (b)?

A: Gingival retraction cord placed carefully into the periodontal pocket, to help to keep the gingival tissue away from the tooth during impression taking. It helps to dry the field and stops bleeding, especially if the margins are subgingival (due to the astringent in which the cord is soaked).